

The project's Kafka topic layer is configured to match how each stream of data is actually used downstream, rather than left on Kafka's defaults. `wearable-vitals`, `smarthome-context`, and `device-connectivity` are all high-frequency, time-ordered event streams — Tier 1 and Tier 2 only ever care about recent readings, so a flat 7-day retention window is enough to support debugging and replay without letting the log grow unbounded. `patient-profile` behaves differently: it's reference data, one static record per patient that occasionally gets updated, so it uses `cleanup.policy=compact` instead of time-based retention — Kafka keeps only the latest message per key (`patient_id`) rather than accumulating every historical update, which is the correct semantics for "current profile," not "profile history." `alarms` sits between the two: it's Tier 1's output and, now, one of Tier 2's inputs, so it's kept a bit longer (30 days) than the raw sensor streams, since alarm history is more valuable to retain for review, evaluation, and the S4–S6 ablation work than a raw vitals reading is. All five topics currently run with a single partition and replication factor of 1, which fits a single-broker local/dev Kafka instance (as in `docker-compose.yml`); this would need to change (more partitions for parallelism, replication factor ≥ 3) before any real multi-broker or production deployment, but for a research prototype it keeps the setup simple and deterministic.

Topic	Key	Retention / Cleanup	Why
<code>wearable-vitals</code>	<code>patient_id</code>	7-day retention	High-frequency vitals stream; only recent history is needed for Tier 1/Tier 2 decisions and replay/debugging
<code>smarthome-context</code>	<code>patient_id</code>	7-day retention	Contextual signal stream (presence, temperature, dose, noise); same recency need as vitals
<code>device-connectivity</code>	<code>device_id</code>	7-day retention	Status stream feeding the S6 confidence-flag logic; only current/recent state matters
<code>patient-profile</code>	<code>patient_id</code>	Compacted (<code>cleanup.policy=compact</code>)	Reference data — one current record per patient, not a history of updates
<code>alarms</code>	<code>patient_id</code>	30-day retention	Tier 1's output and Tier 2's input (for S6); kept longer since alarm history matters for evaluation/ablation, not just live processing

All topics: 1 partition, replication factor 1 — appropriate for the single-broker local dev setup, not intended as a production configuration.

